

INTRODUCTION

The **Cricket Simulator** is an interactive program that allows users to play a virtual cricket match against the computer. Users select their team and an opponent team from a predefined list of international cricket teams. The game involves a toss to decide which team will bat or bowl first, followed by simulated overs where runs and wickets are determined based on user input and random choices by the computer.

Key Features:

1. **Team Selection** : Choose teams from a diverse list of cricket nations.
2. **Toss Mechanism** : Simulates a coin toss to decide the batting or bowling order.
3. **Gameplay** : Players input numbers to simulate runs, with wickets determined by matching inputs.
4. **Overs and Innings** : The game is played over a user-specified number of overs in two innings.
5. **Result Calculation** : Determines the winner based on scores or declares a tie if applicable.
6. **Database Integration** : Match results are saved to a MySQL database for future reference.

SOURCE CODE

```
import random
#import mysql.connector as sql

print("""Welcome to CRICKET SIMULATOR.
The Cricket Simulator is an interactive program that allows users to play a virtual cricket match
against the computer.
Users select their team and an opponent team from a predefined list of international cricket
teams (India, Australia,
England, New Zealand, and Pakistan). The game involves a toss to decide which team will bat
or bowl first, followed by
simulated overs where runs and wickets are determined based on user input and random
choices by the computer.
""")

#Players
print("""
Players:
India=[Rohit Sharma, Shubman Gill, Virat Kohli, Shreyas Iyer, KL Rahul, Ravindra Jadeja,
      Jasprit Bumrah, Hardik Pandya, Kuldeep Yadav, Mohammed Shami, Suryakumar Yadav]

Australia=[Pat Cummins, Travis Head, Glenn Maxwell, Marnus Labuschagne, Steve Smith,
David Warner,
      Mitchell Marsh, Marcus Soinis, Cameron Green, Mitchell Starc, Adam Zampa]

England=[Jos Buttler, Moeen Ali, Jonny Bairstow, Harry Brook, Sam Curran, Liam Livingstone,
      Adil Rashid, Joe Root, Ben Stokes, Mark Wood, Chris Woakes]

New Zealand=[Kane Williamson, Trent Boult, Devon Conway, Lockie Ferguson, Tom Latham,
Daryl Mitchell,
      Jimmy Neesham, Glenn Phillips, Rachin Ravindra, Mitch Santner, Tim Southee]

Pakistan=[Abdulla Shafique, Fakhar Zaman, Babar Azam, Mohammed Rizwan, Saud Shakeel,
Agha Salman,
      Iftikhar Ahmed, Shadab Khan, Shaheen Shah Afridi, Mohammad Wasim,Haris Rauf]""")

ind=['Rohit Sharma','Shubman Gill','Virat Kohli','Shreyas Iyer','KL Rahul','Ravindra Jadeja',
```

'Jasprit Bumrah','Hardik Pandya','Kuldeep Yadav','Mohammed Shami','Suryakumar Yadav']

aus=['Pat Cummins','Travis Head','Glenn Maxwell','Marnus Labuschagne','Steve Smith','David Warner',

'Mitchell Marsh','Marcus Soinis','Cameron Green','Mitchell Starc','Adam Zampa']

eng=['Jos Buttler','Moeen Ali','Jonny Bairstow','Harry Brook','Sam Curran','Liam Livingstone',
'Adil Rashid','Joe Root','Ben Stokes','Mark Wood','Chris Woakes']

nz=['Kane Williamson','Trent Boult','Devon Conway','Lockie Ferguson','Tom Latham','Daryl Mitchell',

'Jimmy Neesham','Glenn Phillips','Rachin Ravindra','Mitch Santner','Tim Southee']

pak=['Abdulla Shafique','Fakhar Zaman','Babar Azam','Mohd Rizwan','Saud Shakeel','Agha Salman',

'Iftikhar Ahmed','Shadab Khan','Shaheen Shah Afridi','Mohammad Wasim','Haris Rauf']

#Team Selection

print("\n

Teams:

1) India

2) Australia

3) England

4) New Zealand

5) Pakistan")

t=input("Select Your Team : ")

t1=t.title()

if t1=='India':

print("India =",ind)

elif t1=='Australia':

print("Australia =",aus)

elif t1=='England':

print("England =",eng)

elif t1=='New Zealand':

print("New Zealand =",nz)

elif t1=='Pakistan':

print("Pakistan =",pak)

else:

print("TEAM NOT FOUND.")

print("\n

Teams:

1) India

```

2) Australia
3) England
4) New Zealand
5) Pakistan")
t=input("Select Opponent's Team : ")
t2=t.title()
if t2=='India':
    print("India =",ind)
elif t2=='Australia':
    print("Australia =",aus)
elif t2=='England':
    print("England =",eng)
elif t2=='New Zealand':
    print("New Zealand =",nz)
elif t2=='Pakistan':
    print("Pakistan =",pak)
else:
    print("TEAM NOT FOUND.")

print("")

#Toss
bat=""
bowl=""
def toss():
    a=input(f"{t1} select Heads or Tails : ")
    coin=('Heads','Tails')
    choice=random.choice(coin)
    if a.title()==choice:
        print(f"{t1} won the toss and elected to ")
        ch=input("Bat or Bowl : ")
        choice=ch.title()
        if choice=="Bat":
            return t1,t2
        elif choice=="Bowl":
            return t2,t1
        else:
            print("INVALID INPUT")

    else:
        ch=('Bat','Bowl')
        choice=random.choice(ch)
        print(f"{t2} won the toss and elected to",choice)
        if choice=="Bat":

```

```
        return t2,t1
    else:
        return t1,t2
```

```
bat,bowl=toss()
```

```
#Batting
print("\nBatting Team =",bat)
print("\nBowling Team =",bowl)
print("\n")
total_score=0
wickets=0
```

```
#Simulate one over
def over(total_score, wickets):
    tr=0
    wkt=0
    if bat==t1:
        run=0
        for b in range(1, 7):
            uc=int(input("Enter number : "))
            cc=random.choice([0, 1, 2, 3, 4, 6])
            print("the computer choice:",cc)
            if uc==cc:
                print("WICKET!")
                wkt+=1
                run=0
            elif uc!=cc:
                if uc in (1,2,3):
                    if uc==cc+1:
                        run=0
                    elif uc==cc-1:
                        run=0
                else:
                    run=uc
                    tr+=run
            elif uc==0:
                if uc==cc:
                    print("WICKET!")
                    wkt+=1
                    run=0
                else:
                    run=cc
                    tr+=run
```

```

elif uc in(4,6):
    if uc==4 and cc==6 or uc==6 and cc==4 or uc==4 and cc==3 or uc==3 and cc==4:
        run=0
    elif uc==cc:
        print("WICKET!")
        wkt+=1
        run=0
    else:
        run=uc
        tr+=run
    print(f"Ball-{b} : {run} runs")
else:
    run=0
    for b in range(1, 7):
        uc=int(input("Enter number : "))
        cc=random.choice([0, 1, 2, 3, 4, 6])
        print("the computer choice:",cc)
        if uc==cc:
            print("WICKET!")
            wkt+=1
            run=0
        elif uc!=cc:
            if cc in(1,2,3):
                if uc==cc+1:
                    run=0
                elif uc==cc-1:
                    run=0
            else:
                run=cc
                tr+=run
        elif cc==0:
            if uc==cc:
                print("WICKET!")
                wkt+=1
                run=0
            else:
                run=uc
                tr+=run
        elif cc in(4,6):
            if uc==4 and cc==6 or uc==6 and cc==4 or uc==4 and cc==3 or uc==3 and cc==4:
                run=0
            elif uc==cc:
                print("WICKET!")
                wkt+=1

```

```

        run=0
    else:
        run=cc
        tr+=run

    print(f"> Ball-{b} : {run} runs")
    print("Wickets this over =",wkt)
    print("Total runs this over =",tr)
    total_score=total_score+tr
    wickets=wickets+wkt
    return total_score, wickets

#No of overs
total_overs=int(input("Enter number of overs to be played : "))

# Overs function
def innings(total_overs, batting_team):
    total_score = 0
    wickets = 0
    for i in range(1, total_overs + 1):
        print(f"\n Over - {i}")
        total_score, wickets = over(total_score, wickets)
    return total_score, wickets

print("\n-----> INNING - 1 <-----")
if t1==bat:
    print("-----> ",t1,"is batting")
    print("-----> ",t2,"is bowling")
elif t2==bat:
    print("-----> ",t2,"is batting")
    print("-----> ",t1,"is bowling")
ts1, w1 = innings(total_overs, bat)
print(f"Total score of {bat}: {ts1} runs with {w1} wickets.")

print("\n-----> INNING - 2 <-----")
bat=bowl
if t1==bat:
    print("-----> ",t1,"is batting")
    print("-----> ",t2,"is bowling")
elif t2==bat:
    print("-----> ",t2,"is batting")
    print("-----> ",t1,"is bowling")
ts2, w2 = innings(total_overs, bowl)
print(f"Total score of {bowl}: {ts2} runs with {w2} wickets.")

```

```

# Final Result
def results():
    print("\n-----> MATCH RESULT <-----")
    if ts1 > ts2:
        res=f"{bat} wins by {ts1 - ts2} runs!"
        print(res)
    elif ts2 > ts1:
        res=f"{bowl} wins by {10 - w2} wickets!"
        print(res)
    else:
        res="It's a tie!"
        print(res)
    return res

result=results()

#Database Integration
con=sql.connect(host='localhost',user='root',password='dlab',database='cricket')
cur=con.cursor()
cur.execute("select sno from scorecard")
sno=cur.fetchall()
a=sno[-1][0]+1
cur.execute("insert into scorecard values(%s,'%s','%s','%s')" %(a,t1,t2,result))
con.commit()
con.close()

```

MySQL Queries :

```
mysql> create database cricket;
```

```
mysql> use cricket;
```

```
mysql> create table scorecard (Sno int primary key, Your_Team varchar(20),
Opponent_Team varchar(20), Result varchar(50));
```


OUTPUTS

Output-1 :

Welcome to CRICKET SIMULATOR.

The Cricket Simulator is an interactive program that allows users to play a virtual cricket match against the computer.

Users select their team and an opponent team from a predefined list of international cricket teams (India, Australia, England, New Zealand, and Pakistan). The game involves a toss to decide which team will bat or bowl first, followed by simulated overs where runs and wickets are determined based on user input and random choices by the computer.

India=[Rohit Sharma, Shubman Gill, Virat Kohli, Shreyas Iyer, KL Rahul, Ravindra Jadeja,
Jasprit Bumrah, Hardik Pandya, Kuldeep Yadav, Mohammed Shami, Suryakumar Yadav]

Australia=[Pat Cummins, Travis Head, Glenn Maxwell, Marnus Labuschagne, Steve Smith, David Warner,
Mitchell Marsh, Marcus Stoinis, Cameron Green, Mitchell Starc, Adam Zampa]

England=[Jos Buttler, Moeen Ali, Jonny Bairstow, Harry Brook, Sam Curran, Liam Livingstone,
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Iftikhar Ahmed, Shadab Khan, Shaheen Shah Afridi, Mohammad Wasim, Haris Rauf]

Teams:

- 1) India
- 2) Australia
- 3) England
- 4) New Zealand
- 5) Pakistan

Select Your Team : INDIA

India = ['Rohit Sharma', 'Shubman Gill', 'Virat Kohli', 'Shreyas Iyer', 'KL Rahul', 'Ravindra Jadeja', 'Jasprit Bumrah', 'Hardik Pandya', 'Kuldeep Yadav', 'Mohammed Shami', 'Suryakumar Yadav']

Teams:

- 1) India
- 2) Australia
- 3) England
- 4) New Zealand
- 5) Pakistan

Select Opponent's Team : NEW ZEALAND

New Zealand = ['Kane Williamson', 'Trent Boult', 'Devon Conway', 'Lockie Ferguson', 'Tom Latham', 'Daryl Mitchell', 'Jimmy Neesham', 'Glenn Phillips', 'Rachin Ravindra', 'Mitch Santner', 'Tim Southee']

India select Heads or Tails : HEADS

New Zealand won the toss and elected to Bowl

Batting Team = India

Bowling Team = New Zealand

Enter number of overs to be played : 1

-----> INNING - 1 <-----

-----> India is batting

-----> New Zealand is bowling

Over - 1

Enter number : 3

the computer choice: 1

Ball-1 : 3 runs

Enter number : 6

the computer choice: 6

WICKET!

Ball-2 : 0 runs

Enter number : 0

the computer choice: 1
Ball-3 : 1 runs
Enter number : 6
the computer choice: 6
WICKET!
Ball-4 : 0 runs
Enter number : 2
the computer choice: 3
Ball-5 : 0 runs
Enter number : 4
the computer choice: 2
Ball-6 : 4 runs
Wickets this over = 2
Total runs this over = 8
Total score of India: 8 runs with 2 wickets.

-----> INNING - 2 <-----
-----> New Zealand is batting
-----> India is bowling

Over - 1
Enter number : 6
the computer choice: 3
> Ball-1 : 3 runs
Enter number : 4
the computer choice: 1
> Ball-2 : 1 runs
Enter number : 4
the computer choice: 0
> Ball-3 : 4 runs
Enter number : 0
the computer choice: 6
> Ball-4 : 6 runs
Enter number : 2
the computer choice: 6
> Ball-5 : 6 runs
Enter number : 6
the computer choice: 4
> Ball-6 : 0 runs
Wickets this over = 0
Total runs this over = 20
Total score of New Zealand: 20 runs with 0 wickets.

-----> MATCH RESULT <-----
New Zealand wins by 10 wickets!

Output-2 :

Welcome to CRICKET SIMULATOR.

The Cricket Simulator is an interactive program that allows users to play a virtual cricket match against the computer.

Users select their team and an opponent team from a predefined list of international cricket teams (India, Australia, England, New Zealand, and Pakistan). The game involves a toss to decide which team will bat or bowl first, followed by simulated overs where runs and wickets are determined based on user input and random choices by the computer.

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Jasprit Bumrah, Hardik Pandya, Kuldeep Yadav, Mohammed Shami, Suryakumar Yadav]

Australia=[Pat Cummins, Travis Head, Glenn Maxwell, Marnus Labuschagne, Steve Smith, David Warner,
Mitchell Marsh, Marcus Stoinis, Cameron Green, Mitchell Starc, Adam Zampa]

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Jimmy Neesham, Glenn Phillips, Rachin Ravindra, Mitch Santner, Tim Southee]

Pakistan=[Abdulla Shafique, Fakhar Zaman, Babar Azam, Mohammed Rizwan, Saud Shakeel, Agha Salman,
Iftikhar Ahmed, Shadab Khan, Shaheen Shah Afridi, Mohammad Wasim, Haris Rauf]

Teams:

- 1) India
- 2) Australia
- 3) England
- 4) New Zealand
- 5) Pakistan

Select Your Team : Australia

Australia = ['Pat Cummins', 'Travis Head', 'Glenn Maxwell', 'Marnus Labuschagne', 'Steve Smith', 'David Warner', 'Mitchell Marsh', 'Marcus Soinis', 'Cameron Green', 'Mitchell Starc', 'Adam Zampa']

Teams:

- 1) India
- 2) Australia
- 3) England
- 4) New Zealand
- 5) Pakistan

Select Opponent's Team : Pakistan

Pakistan = ['Abdulla Shafique', 'Fakhar Zaman', 'Babar Azam', 'Mohd Rizwan', 'Saud Shakeel', 'Agha Salman', 'Iftikhar Ahmed', 'Shadab Khan', 'Shaheen Shah Afridi', 'Mohammad Wasim', 'Haris Rauf']

Australia select Heads or Tails : Tails

Pakistan won the toss and elected to Bat

Batting Team = Pakistan

Bowling Team = Australia

Enter number of overs to be played : 1

-----> INNING - 1 <-----

-----> Pakistan is batting

-----> Australia is bowling

Over - 1

Enter number : 0

the computer choice: 3

> Ball-1 : 3 runs

Enter number : 6

the computer choice: 6

WICKET!

> Ball-2 : 0 runs

Enter number : 1

the computer choice: 2

> Ball-3 : 0 runs

Enter number : 1

the computer choice: 4

> Ball-4 : 4 runs

Enter number : 2

the computer choice: 6

> Ball-5 : 6 runs

Enter number : 6
the computer choice: 6
WICKET!
> Ball-6 : 0 runs
Wickets this over = 2
Total runs this over = 13
Total score of Pakistan: 13 runs with 2 wickets.

-----> INNING - 2 <-----
-----> Australia is batting
-----> Pakistan is bowling

Over - 1
Enter number : 0
the computer choice: 6
Ball-1 : 6 runs
Enter number : 2
the computer choice: 2
WICKET!
Ball-2 : 0 runs
Enter number : 4
the computer choice: 2
Ball-3 : 4 runs
Enter number : 4
the computer choice: 2
Ball-4 : 4 runs
Enter number : 6
the computer choice: 0
Ball-5 : 6 runs
Enter number : 0
the computer choice: 2
Ball-6 : 2 runs
Wickets this over = 1
Total runs this over = 22
Total score of Australia: 22 runs with 1 wickets.

-----> MATCH RESULT <-----
Australia wins by 9 wickets!

Output-3 :

Welcome to CRICKET SIMULATOR.

The Cricket Simulator is an interactive program that allows users to play a virtual cricket match against the computer.

Users select their team and an opponent team from a predefined list of international cricket teams (India, Australia, England, New Zealand, and Pakistan). The game involves a toss to decide which team will bat or bowl first, followed by simulated overs where runs and wickets are determined based on user input and random choices by the computer.

India=[Rohit Sharma, Shubman Gill, Virat Kohli, Shreyas Iyer, KL Rahul, Ravindra Jadeja,
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Australia=[Pat Cummins, Travis Head, Glenn Maxwell, Marnus Labuschagne, Steve Smith, David Warner,
Mitchell Marsh, Marcus Stoinis, Cameron Green, Mitchell Starc, Adam Zampa]

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Adil Rashid, Joe Root, Ben Stokes, Mark Wood, Chris Woakes]

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Jimmy Neesham, Glenn Phillips, Rachin Ravindra, Mitch Santner, Tim Southee]

Pakistan=[Abdulla Shafique, Fakhar Zaman, Babar Azam, Mohammed Rizwan, Saud Shakeel, Agha Salman,
Iftikhar Ahmed, Shadab Khan, Shaheen Shah Afridi, Mohammad Wasim,Haris Rauf]

Teams:

- 1) India
- 2) Australia
- 3) England
- 4) New Zealand

5) Pakistan

Select Your Team : INDIA

India = ['Rohit Sharma', 'Shubman Gill', 'Virat Kohli', 'Shreyas Iyer',
'KL Rahul', 'Ravindra Jadeja', 'Jasprit Bumrah', 'Hardik Pandya', 'Kuldeep
Yadav', 'Mohammed Shami', 'Suryakumar Yadav']

Teams:

1) India

2) Australia

3) England

4) New Zealand

5) Pakistan

Select Opponent's Team : PAKISTAN

Pakistan = ['Abdulla Shafique', 'Fakhar Zaman', 'Babar Azam', 'Mohd
Rizwan', 'Saud Shakeel', 'Agha Salman', 'Iftikhar Ahmed', 'Shadab Khan',
'Shaheen Shah Afridi', 'Mohammad Wasim', 'Haris Rauf']

India select Heads or Tails : TAILS

India won the toss and elected to

Bat or Bowl : BOWL

Batting Team = Pakistan

Bowling Team = India

Enter number of overs to be played : 2

-----> INNING - 1 <-----

-----> Pakistan is batting

-----> India is bowling

Over - 1

Enter number : 4

the computer choice: 6

> Ball-1 : 0 runs

Enter number : 0

the computer choice: 4

> Ball-2 : 4 runs

Enter number : 1

the computer choice: 4

> Ball-3 : 4 runs

Enter number : 6

the computer choice: 3

> Ball-4 : 3 runs

Enter number : 3

the computer choice: 4


```
> Ball-5 : 0 runs
Enter number : 2
the computer choice: 6
> Ball-6 : 6 runs
Wickets this over = 0
Total runs this over = 17
```

```
Over - 2
Enter number : 2
the computer choice: 1
> Ball-1 : 0 runs
Enter number : 4
the computer choice: 6
> Ball-2 : 0 runs
Enter number : 3
the computer choice: 4
> Ball-3 : 0 runs
Enter number : 0
the computer choice: 0
WICKET!
> Ball-4 : 0 runs
Enter number : 0
the computer choice: 1
> Ball-5 : 0 runs
Enter number : 1
the computer choice: 0
> Ball-6 : 1 runs
Wickets this over = 1
Total runs this over = 1
Total score of Pakistan: 18 runs with 1 wickets.
```

```
-----> INNING - 2 <-----
-----> India is batting
-----> Pakistan is bowling
```

```
Over - 1
Enter number : 6
the computer choice: 6
WICKET!
Ball-1 : 0 runs
Enter number : 6
the computer choice: 6
WICKET!
Ball-2 : 0 runs
Enter number : 6
the computer choice: 1
Ball-3 : 6 runs
```

Enter number : 6
the computer choice: 2
Ball-4 : 6 runs
Enter number : 6
the computer choice: 0
Ball-5 : 6 runs
Enter number : 6
the computer choice: 3
Ball-6 : 6 runs
Wickets this over = 2
Total runs this over = 24

Over - 2
Enter number : 3
the computer choice: 1
Ball-1 : 3 runs
Enter number : 4
the computer choice: 3
Ball-2 : 0 runs
Enter number : 6
the computer choice: 1
Ball-3 : 6 runs
Enter number : 0
the computer choice: 0
WICKET!
Ball-4 : 0 runs
Enter number : 6
the computer choice: 0
Ball-5 : 6 runs
Enter number : 3
the computer choice: 1
Ball-6 : 3 runs
Wickets this over = 1
Total runs this over = 18
Total score of India: 42 runs with 3 wickets.

-----> MATCH RESULT <-----
India wins by 7 wickets!

MySQL Output :

```
mysql> select * from scorecard;
```

sno	Your_Team	Opponent_Team	Results
1	India	Pakistan	India wins by 9 wickets!
2	Australia	India	India wins by 23 runs!
3	India	Pakistan	Pakistan wins by 15 runs!
4	Australia	England	England wins by 3 runs!
5	India	New Zealand	New Zealand wins by 10 wickets!
6	Australia	Pakistan	Australia wins by 9 wickets!
7	India	Pakistan	India wins by 7 wickets!
8	India	England	England wins by 22 runs!
9	India	Australia	Australia wins by 108 runs!
10	England	Pakistan	Pakistan wins by 8 wickets!

```
10 rows in set (0.00 sec)
```

CONCLUSION

The **Cricket Simulator** is a captivating program that blends user interaction, strategy, and random chance to replicate the excitement of a cricket match. By allowing players to select teams, make crucial gameplay decisions, and compete against the computer, it provides a highly engaging experience for cricket enthusiasts.

The integration of database functionality ensures match results are preserved, adding a layer of authenticity and allowing users to revisit their gaming history. Whether you're a die-hard cricket fan or someone looking for a fun simulation, this program delivers a thrilling virtual match experience.

The Cricket Simulator exemplifies how technology can bring the spirit of sports into a digital environment, offering both entertainment and a touch of competitive excitement.